

$$\text{Fm}_R(\text{RelIdx}) \xrightarrow{\text{rel}_\rho} \text{Fm}_R(\kappa) \xrightarrow{\text{ev}_\rho} V(\kappa)$$

$$\text{range}(\text{rel}_\rho) = \ker(\text{ev}_\rho)$$

$$\text{given } w = \sum_e a_e \delta_e \in \ker(\text{ev}_\rho), \text{ let } q_\rho(w) = \sum_e a_e \delta_{\text{const}(\text{ev}_\rho e)}.$$

① **derivation certificate**

$$\text{rel}_\rho(c_{\text{der}}) = w - q_\rho(w)$$

each expression is first read back
by its own normalization derivation.

② **scalar certificate**

$$\text{rel}_\rho(c_{\text{scl}}) = q_\rho(w)$$

the remaining formal sum is caught by
the addition and scalar module laws.

$$\text{rel}_\rho(c_{\text{der}} + c_{\text{scl}}) = w$$

the two families of module laws: $\delta_v + \delta_u - \delta_{v+u}$ and $\delta_{av} - a\delta_v$; every relation is erased by evaluation.